

NETWORK INTRUSION DETECTION SYSTEM

Using Random Forest + Deep Neural Networks

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Introduction

Why intelligent intrusion detection is needed

- Computer networks are increasingly used in banking, healthcare, education, business and communication.
- Growing connectivity increases exposure to cyberattacks that can compromise data and disrupt services.
- A Network Intrusion Detection System (NIDS) monitors network traffic and identifies suspicious activity.
- Machine Learning can learn traffic patterns and classify records as normal or malicious.
- Our project implements Random Forest and Deep Neural Network models with an interactive detection dashboard.

Problem Statement & Objectives

Problem Statement

Traditional intrusion detection approaches may struggle with complex traffic patterns. The project builds an intelligent system that learns from labeled network traffic and automatically identifies suspicious activity.

Objectives

- Detect malicious network traffic
- Perform preprocessing and feature engineering
- Train Random Forest and DNN models
- Compare Accuracy, Precision, Recall and F1-Score
- Combine predictions using a hybrid approach
- Generate threat levels, confidence and alerts

From network traffic to actionable threat information

Literature Survey

Selected studies from the project's literature review

Year	Study	Dataset	Method
2016	Random Forest Modeling for NIDS	NSL-KDD	Random Forest
2018	Performance Comparison of SVM, RF & ELM	NSL-KDD	SVM, RF, ELM
2019	Network Intrusion Detection Algorithm	KDD99 / NSL-KDD	DNN
2020	Deep Learning for Cyber Security IDS	CSE-CIC-IDS2018 / Bot-IoT	DNN, RNN, CNN, RF
2022	Deep Stacking Network for IDS	NSL-KDD	DNN, RF, Deep Stacking
2024	NIDS Leveraging Multimodal Features	UNSW-NB15	RF + Soft Voting

Dataset Description

Dataset

Prepared network intrusion dataset
2,000 records
44 original input features

Traffic Classes

Normal
DoS
Probe
R2L
U2R

Example Features

duration, protocol_type,
service, flag
src_bytes, dst_bytes, count
error and host features

Engineered Features

Traffic Volume
Upload/Download Ratio
Connection Intensity
Service Error Score
Service Consistency

Behavioral Features

Host Anomaly Score
Privilege Activity
Login Risk
Threat Behavior Score
Threat Level

Network traffic records used for supervised learning

Exploratory Data Analysis

EDA Checks

- Dataset shape and columns
- Missing-value analysis
- Attack-type distribution
- Threat-level distribution

Behavioral Analysis

- Traffic volume
- Service errors
- Host anomaly behavior
- Login and privilege activity

Graphs to Insert

Attack Distribution • Threat Level Distribution • Threat Behavior Score • RF vs DNN Performance • Confusion Matrix

Understanding the traffic before model training

Methodology

1

Dataset

2

Preprocessing

3

Feature Engineering

4

Train/Test Split

5

Random Forest

6

DNN

7

Hybrid Prediction

8

Threat Analysis

9

Dashboard

End-to-end machine learning pipeline

Machine Learning Models

Random Forest

- Ensemble of decision trees
- Combines predictions from multiple trees
- Handles mixed network features effectively
- Used for robust feature-based classification
- Produces prediction probabilities for hybrid analysis

Deep Neural Network

- Learns complex feature relationships
- Dense layers: 128 → 64 → 32 neurons
- Dropout for regularization
- Adam optimizer
- Sigmoid output for binary attack classification

Two complementary learning approaches

Evaluation Metrics

Performance is evaluated using multiple measures

Accuracy

Overall proportion of correctly classified records.

Precision

How many predicted attacks are actually attacks.

Recall

How many actual attacks are correctly detected.

F1-Score

Balance between Precision and Recall.

ROC-AUC

Measures class-separation capability across thresholds.

Confusion Matrix

Correct vs incorrect classifications.

Results & Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	99.25%	99.07%	100%	99.53%	100%
DNN	99.75%	100%	99.69%	99.84%	100%
Hybrid RF+DNN	100%	100%	100%	100%	100%

Result Interpretation

Compare models using Accuracy, Precision, Recall, F1-Score and ROC-AUC. Insert the final notebook performance graph above or beside this table.

Replace placeholders with exact notebook values

Hybrid Prediction & Confusion Matrix

Random Forest

Prediction probability

DNN

Prediction probability

Hybrid

Combined prediction

Hybrid Output

- Hybrid prediction
- Hybrid confidence
- Model agreement
- Threat behavior score
- Threat level

Confusion Matrix

Insert the final confusion-matrix screenshot from the notebook here and discuss correct and incorrect classifications.

Combining model outputs for the final decision

Project Output Demonstration

1. Upload

Analyze Network Traffic

Upload your network traffic CSV file

Choose File ML_Networ..._Aligned

2. Analyze

Analyze Network Traffic

Upload your network traffic CSV file

Choose File ML_Networ..._Aligned.csv

ANALYZE NETWORK TRAFFIC

Analyzing network traffic...

3. View Results

Intrusions

1599

Normal Traffic

401

Prediction	Threat Score	Threat Level	Confidence
U2R	21.68	LOW	97.70%
Normal	5.11	LOW	100.00%
R2L	16.94	LOW	71.66%

Conclusion & Future Scope

Summary of the developed Network Intrusion Detection System

- Developed a machine-learning-based NIDS using Random Forest and Deep Neural Network models.
- Performed preprocessing, feature engineering, threat scoring and model evaluation.
- Implemented hybrid RF+DNN prediction and generated threat levels, confidence and security alerts.
- Connected the trained ML system to an interactive frontend for CSV-based network traffic analysis.
- Future scope: real-time packet monitoring, larger datasets, hyperparameter optimization and cloud deployment.

THANK YOU

Questions?